TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

# AI.211

## Pattern Recognition

### COURSE OVERVIEW

Classifying and clustering patterns: feature extraction, classifiers, and evaluation for perception and diagnostics.

PATTERN RECOGNITION

1

SESSION

60 MINUTES

## ORIENTATION: SEEING PATTERNS

**KEY TOPICS**

- What is pattern recognition
- Feature extraction and selection
- Supervised vs. unsupervised
- Applications

**TEACHING FOCUS**

Recognize structure in data.

**KEY TAKEAWAY** Patterns are the signature of meaning.

PATTERN RECOGNITION

2

SESSION

60 MINUTES

## CLASSICAL CLASSIFIERS

**KEY TOPICS**

- Nearest neighbor
- Linear discriminant analysis
- Naive Bayes
- Support vector machines

**TEACHING FOCUS**

Assign patterns to categories.

**KEY TAKEAWAY** Classifiers draw boundaries in feature space.

3

SESSION

60 MINUTES

## FEATURE ENGINEERING

**KEY TOPICS**

- Descriptors and histograms
- Dimensionality reduction
- Feature selection
- Domain-specific features

**TEACHING FOCUS**

Represent patterns compactly.

**KEY TAKEAWAY** Good features make classification easy.

PATTERN RECOGNITION

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SESSION

60 MINUTES

## CLUSTERING

**KEY TOPICS**

- k-means and variants
- Gaussian mixture models
- DBSCAN
- Cluster evaluation

**TEACHING FOCUS**

Group similar patterns together.

**KEY TAKEAWAY** Clustering finds categories without labels.

PATTERN RECOGNITION

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SESSION

60 MINUTES

## EVALUATION

### KEY TOPICS

- Confusion matrix
- ROC and AUC
- Cross-validation
- Precision-recall

### TEACHING FOCUS

Judge a recognizer honestly.

**KEY TAKEAWAY** Accuracy alone can mislead.

PATTERN RECOGNITION

6

SESSION

60 MINUTES

## PATTERN RECOGNITION IN AI & REVIEW

**KEY TOPICS**

- Speech, vision, and sensor patterns
- Course recap and assessment
- Toward robot kinematics (AI.212)
- Recognizer demo

**TEACHING FOCUS**

Build a pattern recognizer end-to-end.

**KEY TAKEAWAY** Recognition turns raw data into categories.